

Hamza Akram

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Professional Summary

Software Engineering postgraduate with practical experience designing, building, testing and maintaining applications. I work mainly in Python, with SQL, debugging, database management and applied AI, on a solid grounding in object oriented programming, data structures, algorithms and software engineering practice. Recently worked remotely as a Generalist Expert with Mercor, evaluating large language model outputs for factual grounding, relevance and reasoning quality against source material. Independently designed and built three privacy focused applications, Aura, TruthLens and Verity, taking each one from architecture through to deployment. Studying for an MSc in Software Engineering at the University of Hertfordshire and eligible to work in the UK.

Skills

- **Programming Languages:** Python, JavaScript, TypeScript, SQL, HTML5, CSS3
- **Artificial Intelligence and Applied AI:** Large Language Models (LLMs), Retrieval Augmented Generation (RAG), evaluating AI model output, prompt engineering, local LLM integration (Ollama)
- **Software Engineering:** Object oriented programming, data structures and algorithms, software design patterns
- **Web Development:** React.js, Node.js, FastAPI, REST API development, frontend and backend development, integrating third party APIs
- **Cloud and Tools:** AWS (EC2, Lambda, S3, SQS, SNS, CloudWatch, IAM, VPC), Git, GitHub, Visual Studio Code, Postman
- **Networking:** TCP/IP, DNS, HTTP/HTTPS, routing, firewalls, load balancing
- **Systems Administration:** Linux and Windows Server administration
- **Technical Support and Communication:** Diagnosing complex technical issues and explaining solutions clearly to people of varying technical background
- **Soft Skills:** Problem solving, analytical thinking, technical communication, team collaboration, adaptability

Projects

Verity: Personal AI Assistant with Verified Research github.com/hamzaakrm/Verity

- Merged two earlier solo projects, Aura and TruthLens, into one application that combines persistent conversational memory with a research pipeline that filters sources for credibility, then extended it with capability neither project had alone.
- Built a routing layer that classifies every message and sends it either to local conversational chat or to the full research pipeline, with manual override.
- Implemented conversational research, so follow up questions are answered from sources already retrieved, cutting response time from minutes to seconds. When the retrieved sources do not contain the answer, the system says so rather than inferring one.
- Designed a dual provider architecture, using local Ollama models or the Gemini and Claude APIs behind a single toggle, so the user can trade privacy for reasoning power while the memory database, web search and source filtering stay on their own machine.
- Added image verification combining EXIF metadata inspection, a vision model score for the likelihood of AI generation, and a search for where the image originated, along with voice conversation in the browser.

Aura: Offline Personal AI Assistant github.com/hamzaakrm/aura

- Built a fully offline desktop AI assistant in Python that runs entirely on the user machine, so no personal data ever leaves it, applying privacy and security by design throughout.
- Integrated local LLMs through Ollama for offline conversation, with memory that persists across sessions, automatic summarisation of older history, and offline voice input and output in several languages.
- Designed a desktop interface in PySide6 backed by SQLite for local storage, with file and application automation and an optional learning mode for web research and content tasks.

TruthLens: Fact Checking and Source Verification Tool github.com/hamzaakrm/TruthLens

- Developed a privacy focused research tool with a FastAPI backend and a local LLM through Ollama that searches the web, cross references several sources, and produces cited answers while flagging uncertainty and disagreement between sources.
- Implemented similarity clustering to detect republished wire stories, so ten outlets carrying one report are not counted as ten independent confirmations, along with automated classification of reporting versus opinion and Wayback Machine checks for sources edited after publication.
- Built local source reputation tracking in SQLite, image verification through EXIF metadata and signals of AI generation, and exportable Markdown and PDF reports with cross referencing across languages.

Work Experience

Generalist Expert, AI Model Evaluation, Mercor (Remote, Contract)

5 August 2026 to 30 September 2026

- Evaluate large language model outputs against source material, assessing claim by claim whether a statement is substantiated or only sounds precise.
- Identify unsupported causal attributions and fluent filler that reads as insight but adds no information beyond the given context.
- Judge relevance in context, where identical phrasing can be substantive in one position of a document and redundant in another.
- Apply grading criteria consistently across a high volume of independent judgments, including borderline cases that resist an easy call.
- Produce structured written justifications for each judgment to support reviewer consistency and downstream data quality.

Developer, Triolinks Services (Pakistan)

March 2024 to June 2025

- Developed and maintained applications in Python supporting business and healthcare management processes.
- Designed and implemented backend functionality in Python, improving application performance and reliability.
- Worked with SQL databases to store, retrieve and manage application data efficiently.
- Helped integrate AI features to improve automation and user experience.
- Took part in software testing, debugging and troubleshooting to keep deliverables reliable.
- Worked with senior developers across the Software Development Life Cycle, including requirements gathering, development, testing and deployment.
- Used Git for version control and team collaboration.
- Prepared technical documentation and contributed to project planning discussions.

Education

University of Hertfordshire

MSc Software Engineering

Sept 2025 to Sept 2027

- Measure and Models for Software Engineering (software metrics, UML modelling, system analysis and design, requirements engineering, process modelling, quality assurance, software performance evaluation, risk analysis, software process improvement)
- Software Engineering Practice and Experience (agile development, team collaboration, Software Development Life Cycle, version control using Git, software testing, code reviews, requirements analysis, project delivery)
- Programming for Software Engineers
- Team Research and Development Project (group software development project, collaborative research, agile methodology, requirement analysis, system design, implementation and testing, version control using Git, final project delivery and presentation)
- Responsible Technology (legal, social, ethical and professional issues)

University of Haripur

BSc Software Engineering (Hons)

February 2020 to February 2024

- Object oriented programming (Java, Python, OOP principles, design patterns, application development)
- Software architecture and design (system design, architectural patterns, UML modelling, scalable software solutions)
- Database systems (SQL, database design, data modelling, query optimisation, database management)
- Web development technologies (HTML, CSS, JavaScript, frontend and backend development)
- Artificial intelligence and machine learning (Python, Anaconda, Jupyter Notebook, algorithms)
- Cloud computing (cloud architecture, AWS services, virtualisation, cloud deployment and management)
- Cyber security fundamentals (network security, risk assessment, security controls, data protection)
- Data structures and algorithms (algorithm design, problem solving, data management)
- Software project management (risk management, agile project delivery, team collaboration)
- Research methods for computing
- Operating systems (Linux, Windows, memory management, system administration)